

04 | OVERVIEW



Bridging the Gap: Climate Data and Community Water Resilience

Abstract

Flooding is the costliest type of disaster in the United States, and climate change is making it more frequent and severe. NOAA's National Water Model can forecast river and stream conditions across the country, but few community planners use it. Researchers interviewed 95 people in seven U.S. communities to learn why, and worked with them to recommend how climate data tools can become more useful for the people planning for floods.

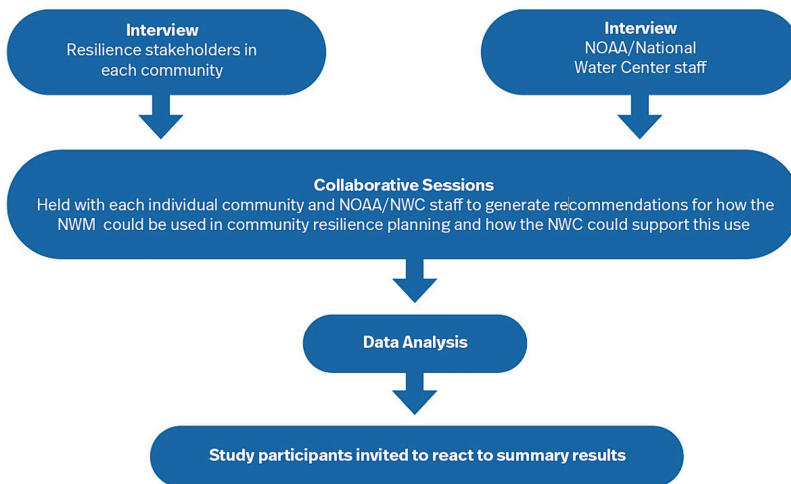
Introduction

As global temperatures rise, inland and coastal flooding are projected to grow more severe, frequent, and intense. Many cities now practice resilience planning: preparing so they can keep critical services running after a disaster and reduce harm from future ones. To plan well, they need good data. Climate services are tools that turn climate science into practical information. One important example is the National Water Model (NWM), run by NOAA's National Water Center. It tracks and predicts precipitation, snowmelt, soil moisture, and streamflow, producing forecasts from 3 hours to 30 days ahead for 2.7 million river reaches. Yet tools like the NWM are often underused or used incorrectly. This study asked how the model could better serve communities facing water disasters.

04 | METHODS

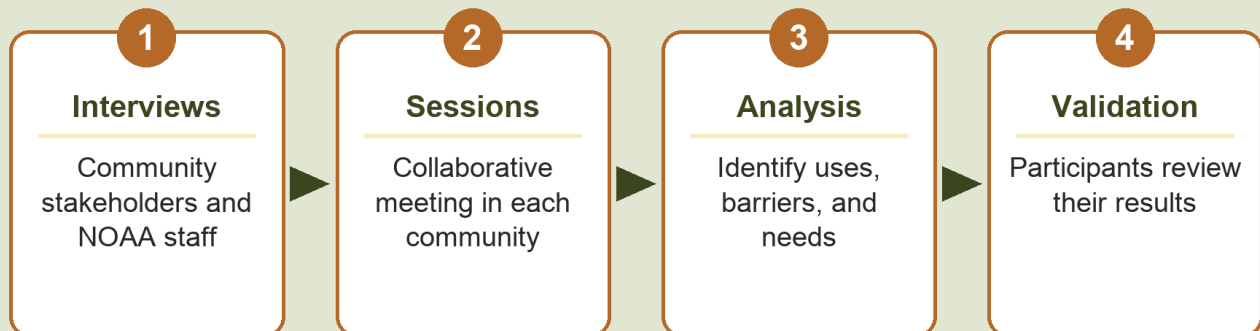
How Do Scientists Study This?

From 2021 to 2023, researchers from Northeastern University's Global Resilience Institute worked with seven geographically diverse communities: Houston, Texas; Burlington, Vermont; Cincinnati, Ohio; Portland, Oregon; Charlotte, North Carolina; Boulder, Colorado; and Minneapolis, Minnesota. Each had faced a water-related hazard in the past five years. The team followed four steps. First, they interviewed community resilience stakeholders and NOAA staff online. Second, they held collaborative sessions in each community. Third, they analyzed the interview and session data. Fourth, they invited participants to review their community's summary results to check that the findings were accurate. This co-development approach placed end-users alongside model developers.


FROM THE ARTICLE

The study's four-step methodology: interviews, collaborative sessions, data analysis, and participant review. Raub et al., 2024, *Frontiers in Climate*, CC BY.

How the study co-developed recommendations



Study the Process

Interview

Researchers interviewed 95 community stakeholders and 14 NOAA National Water Center staff about water hazards and data needs.

Collaborate

Each community held a session with NOAA staff to discuss how the model could support local resilience planning.

Validate

Participants reviewed summaries of their community's results to confirm the findings reflected their views.

What Did They Discover?

Each community faced different water hazards: flat terrain in Houston and Burlington, river confluences in Cincinnati, flash floods and wildfire in Boulder, and sinkhole-prone karst geology under Minneapolis. Across all seven, the researchers found three main barriers: many planners were unaware of the NWM, found it hard to access and interpret, or needed features it did not yet offer. The model proved most useful for shorter timescales, pointing to a wider need for climate services that cover near-, medium-, and long-term planning. Importantly, most people using water data in resilience planning were not trained hydrologists. This shows why tool developers must co-develop with a wide range of users.

The study by the numbers

7

U.S. communities studied from 2021 to 2023

95

community stakeholders interviewed

2.7M

river reaches covered by the National Water Model

The overall interview response rate was 28%. Charlotte's rate was only 9%, so the researchers cautioned against drawing strong conclusions for that city.

04 | TAKEAWAYS

Takeaways

Data needs users

Even a powerful forecast model helps little if planners do not know about it or cannot use it.

Planners are diverse

Most people using water data are not hydrologists, so tools must be easy to access and interpret.

Build it together

Co-developing tools with communities makes climate services more useful and more widely adopted.

Key Terms

Climate services

Tools and information products that translate climate science into practical guidance for decision-makers.

National Water Model (NWM)

A NOAA computer model that tracks and forecasts water across the continental United States, including streamflow, snowmelt, and soil moisture.

Resilience planning

Planning that helps a community keep critical services running after a disaster and reduce harm from future hazards.

Streamflow

The volume of water moving through a river or stream over time.

Co-development

A process in which developers and end-users work together to design a tool so it meets real needs.

04 | ACTIVITY

Activity

Design a Flood Dashboard

Students act as a co-development team and sketch a flood forecast tool that a non-scientist city planner could use.

Materials: Chart paper or a slide template; Markers; Sample river forecast graph (optional)

1. Assign roles: model developer, city planner, emergency manager, and resident.
2. Each role lists the one question they most need the forecast to answer and the time frame they care about.
3. Sketch a one-screen dashboard that answers those questions using simple maps, colors, and plain language.
4. Check your design against the three barriers from the study: awareness, accessibility, and missing capabilities.
5. Present your dashboard and explain one feature that came directly from a teammate's role.

References

Raub, K. B., Laufer, J., Flynn, S. E., Daniels, S., and Sivalingam, T. (2024). Harnessing climate services to support community resilience planning: lessons learned from a community-engaged approach to assessing NOAA's National Water Model. *Frontiers in Climate*, 6.
<https://doi.org/10.3389/fclim.2024.1291165>